245492

Course Name: Database Management System

Course Outcome

CO1-Understand the concept of database management systems and Relational database.
CO2- Identify the various data model used in database design.
CO3- Design conceptual models of a database using ER modeling for real life applications and construct queries in Relational Algebra.
CO4- Create and populate a RDBMS for a real-life application, with constraints and keys using SQL.
CO5- Select the information from a database by formulating complex queries in SQL.
CO6- Analyze the existing design of a database schema and apply concepts of normalization to design an optimal database.
CO7- Discuss indexing mechanisms for efficient retrieval of information from a database.
CO8- Discuss recovery system and be familiar with introduction to web database, distributed databases.

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Mid Term Examination, Even Semester 2025-26
B.Tech. CS/EC(CS)/ECSE/ ME(CS), Year: II/III, Semester: IV/VI
BCSC 1003, Database Management System

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:
1. Attempt all sections
2. No B sheet will be given

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	A hospital stores patient records in files. Doctors, nurses, and billing staff access the same data. Sometimes unauthorized staff view sensitive medical history. How does a DBMS solve this problem better than a file system?	3	1	U	F
2	A company uses a database system to store customer and transaction information. Over time, the database administrator modifies the way data is physically stored on disk and later changes the structure of some tables to support new application requirements. Despite these changes, the existing applications continue to work without any modifications. Based on this scenario, what is data independence? Explain the two types of data independence physical data independence and logical data independence with suitable examples.	3	1	U	F

3	An organization stores employee and project details in a single database table, which can lead to anomalies due to poor design. An insert anomaly occurs when a new project cannot be added unless an employee is assigned to it. An update anomaly arises when changing project information requires updating multiple records, risking inconsistency. A delete anomaly happens when removing the last employee associated with a project also deletes the project's information.	3	2	R	F
4	Let Relation R = (A, B, C, D, E) be a relation scheme with the following Functional dependencies- $AB \rightarrow C$ $C \rightarrow D$ $B \rightarrow E$ Determine all possible candidate keys and how many super keys are possible.	3	3	U	F
5	Consider a university database for the scheduling of classrooms for -final exams. This database could be modeled as the single entity set exam, with attributes course-name, section number, room-number, and time. Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the exam entity set, as: • course with attributes name, department, and c-number • section with attributes s-number and enrollment, and dependent as a weak entity set on course • room with attributes r-number, capacity, and building. create an E-R diagram illustrating the use of all three additional entity sets listed.	3	4	E	C

Section – B

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Suppose a relational schema R = (P, Q, R, S), and set of functional dependency as following: F : { P→ QR, Q →R, P →Q, PQ →R } Find the canonical cover $\mathbf{F_c}$ (Minimal set of functional dependency).	5	6	An	P
7	Explain Domain Constraint, Entity Integrity Constraint & Referential Integrity Constraints? Why each of these is important in a database.	5	4	U	P
8	Consider the following relation schema: Works (Pname,Cname,salary) Lives (Pname,Street,City) Located_in (Cname, city) Manager (Pname, Mgrname) Write the relational algebra queries for the following: a). Find the names of all persons who live in the city Bangalore. b). Retrieve the names of all person of "Infosys" whose salary is between Rs. 50000. c). Find the names of all persons who lives and work in the same city. d). List the names of the people who work for "Tech M" along with the cities they live in. e). Find the average salary of "Infosys" persons.	5	5	An	P